YANQI SU

Computer Science and Software Engineering, The University of Western Australia
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RESEARCH INTERESTS

My research lies at the intersection of Software Engineering, Knowledge Graphs, and Artificial Intelligence, focusing on:

- **Exploratory & change-aware testing**: software and agentic systems
- **Agent skills**: static analysis and testing
- **Agent networks**: supporting their development and deployment, e.g., implementing, maintaining, and testing protocols such as A2A
- **Mining software repositories**: bug triaging and software maintenance

EDUCATION

The Australian National University (ANU) *Sep. 2020 - Sep. 2024*
Ph.D. in Computer Science
Supervisor: Zhenchang Xing

Nanjing University of Aeronautics and Astronautics (NUAA) *Sep. 2017 - Jun. 2020*
M.E. in Computer Science

Nanjing University of Aeronautics and Astronautics (NUAA) *Sep. 2013 - Jun. 2017*
B.E. in Computer Science

PUBLICATIONS

Yanqi Su, Michael Pradel, Chunyang Chen: RippleGUItester: Change-Aware Exploratory Testing. The 35th SIGSOFT International Symposium on Software Testing and Analysis (ISSTA 2026). Acceptance rate: 10.1% (90/888)

Yanqi Su, Zhenchang Xing, Chong Wang, Chunyang Chen, Xiwei Xu, Qinghua Lu, Liming Zhu: Automated Soap Opera Testing Directed by LLMs and Scenario Knowledge: Feasibility, Challenges, and Road Ahead. The ACM International Conference on the Foundations of Software Engineering (FSE2025).

Yanqi Su, Dianshu Liao, Zhenchang Xing, Qing Huang, Mulong Xie, Qinghua Lu, Xiwei Xu: Enhancing Exploratory Testing by Large Language Model and Knowledge Graph. The 46th International Conference on Software Engineering (ICSE'24).

Yanqi Su, Zheming Han, Zhenchang Xing, Xiwei Xu, Liming Zhu, Qinghua Lu: SoapOperaTG: A Tool for System Knowledge Graph Based Soap Opera Test Generation. The 45th International Conference on Software Engineering (ICSE'23 Tool Demo).

Yanqi Su, Zheming Han, Zhenchang Xing, Xin Xia, Xiwei Xu, Liming Zhu, Qinghua Lu: Constructing a System Knowledge Graph of User Tasks and Failures from Bug Reports to Support Soap Opera Testing. 37th IEEE/ACM International Conference on Automated Software Engineering (ASE2022).

Yanqi Su, Zhenchang Xing, Xin Peng, Xin Xia, Chong Wang, Xiwei Xu, Liming Zhu: Reducing Bug

Triaging Confusion by Learning from Mistakes with a Bug Tossing Knowledge Graph. 36th IEEE/ACM International Conference on Automated Software Engineering (ASE2021).
Distinguished Paper Award

Yanqi Su, Zheming Han, Zhipeng Gao, Zhenchang Xing, Qinghua Lu, Xiwei Xu: Still Confusing for Bug-Component Triaging? Deep Feature Learning and Ensemble Setting to Rescue. The 31st IEEE/ACM International Conference on Program Comprehension (ICPC2023).

Yu Zhou, **Yanqi Su**, Taolue Chen, Zhiqiu Huang, Harald C. Gall, Sebastiano Panichella: User Review-Based Change File Localization for Mobile Applications. IEEE Trans. on Software Engineering. (TSE)

Zhipeng Gao, **Yanqi Su**, Xing Hu, Xin Xia: Automating TODO-dropped Methods Detection and Patching. ACM Transactions on Software Engineering and Methodology. (TOSEM)

TEACHING AND WORKING

Lecturer in The University of Western Australia (UWA), Computer Science and Software Engineering, from May 2026

Lecturer in UWA, CITS5501 / CITS3501 Software Testing and Quality Assurance, 2026 Semester 2

Postdoctoral Researcher in Technische Universität München (TUM), Software Engineering & AI, from Oct-21-2024 to Apr-2026, supervised by Chunyang Chen

Lecturer in TUM, Seminar Exploratory Software Testing (INHN0015, INHN4016, IN0014, IN45111), 2026 Summer

Lecturer in TUM, Seminar Exploratory Software Testing (IN0014) (INHN0015, INHN4012), 2025 Summer

Guest Lecturer in TUM, Advanced Topics in Software Engineering (CITHN3003), 2024 & 2025 Winter

Teaching Assistant in ANU, COMP4130 Managing Software Quality and Process, 2024S1, supervised by Xiaoyu Sun

Research Visit in University of Stuttgart & CISPA, Software Lab, from Nov-01-2025 to Nov-12-2025, supervised by Michael Pradel

Research Assistant in Fudan University, CodeWisdom Team, from Sep-01-2020 to Jun-30-2021, supervised by Xin Peng

Teaching Assistant in NUAA, Software Metrics, 2017S2, supervised by Yu Zhou

MENTOR

2025 Abdelrahman Diab, TUM Master thesis: Bug-Inducing Code Change Detection

2025 Lazar Minic, TUM Master thesis: Bridging Policy and Practice: Leveraging LLMs for Privacy-Aware Exploratory Software Testing

2024 Alper Temel, TUM Master thesis: Non-intrusive GUI Element Location Using LLMs

2024 Linsen Mao, TUM Bachelor thesis: Soap Opera Testing on Enterprise Resource Planning Software

2024 Bryan Alvin, TUM Bachelor thesis: Automated Exploratory Testing of Business Logic in ERP Systems

2022 Zheming Han, ANU Master thesis: Reducing Bug Component Triaging Mistakes by Deep Learning

2017 Jian Liu, NUAA Bachelor thesis: Code Reviewer Recommendation Based on File Path Analysis

FUNDINGS & AWARDS

ANU Vice-Chancellor's Travel Grant

ACM SIGSOFT CAPS AWARD FSE/ISSTA 2025

ACM Travel Grant for 2024 IEEE/ACM 46th International Conference on Software Engineering

ACM SIGSOFT Distinguished Paper Award (First author) in ASE2021

HDR Fee Remission Merit Scholarship, ANU, 2020.09 – 2024.09

University Research Scholarship, ANU, 2020.09 – 2024.09

First-Class Graduate Scholarship, NUAA, 2017.09 – 2020.04

Outstanding Research and Innovation Award, NUAA, 2019.09 – 2020.04

Outstanding Contribution to Social Activities, NUAA, 2019.09 – 2020.04

Third-Class Scholarship, NUAA, 2015.09 – 2017.06

Second-Class Scholarship, NUAA, 2014.09 – 2015.06

Third-Class Scholarship, NUAA, 2013.09 – 2014.06

Microsoft Research Asia Club Star, 2014

Outstanding Youth Volunteer, 2nd Youth Olympic Games, 2014.08 – 2014.09

SERVICE

Journal Reviewer

- IEEE Transactions on Software Engineering (TSE)
- ACM Transactions on Software Engineering and Methodology (TOSEM)
- Empirical Software Engineering (EMSE)
- Software Testing, Verification and Reliability (STVR)

Conference Program Committee

- The ACM International Conference on the Foundations of Software Engineering (FSE2027)
- The 49th IEEE/ACM International Conference on Software Engineering (ICSE2027)
- The 41th IEEE/ACM International Conference on Automated Software Engineering (ASE2026)

- The 37th IEEE International Symposium on Software Reliability Engineering (ISSRE2026)
- The 23rd International Conference on Mining Software Repositories (MSR2026)
- The 17th International Conference on Internetware (Internetware 2026)
- The International Conference on Predictive Models and Data Analytics in Software Engineering (PROMISE 2026)
- The 1st International Workshop on User Interface and Experience for Software Engineering (UISE 2026)
- The 40th IEEE/ACM International Conference on Automated Software Engineering (ASE2025)
- The 36th IEEE International Symposium on Software Reliability Engineering (ISSRE2025)
- The 32nd Asia-Pacific Software Engineering Conference (APSEC2025 SEIP Track & Student Research Competition Track)

Student Volunteer

- The ACM International Conference on the Foundations of Software Engineering (FSE2025)

TALK

Shaping the Future of Exploratory Software Testing with Knowledge Graphs and LLMs
This talk is a compilation of my main research and I gave this talk at multiple places.

- Jul. 2025 The Chinese University of Hong Kong, Shenzhen, China
- Jun. 2025 University of Oxford, UK
- Apr. 2025 Aalto University, Finland

Automated Soap Opera Testing Directed by LLMs and Scenario Knowledge

- Feb. 2025 Technical University of Munich, Germany

Constructing a System Knowledge Graph of User Tasks and Failures to Support Soap Opera Testing

- Dec. 2023 Google Research Australia, Australia

REFERENCES

Zhenchang Xing (Supervisor)
Associate Professor
School of Computing
The Australian National University
Senior Principal Research Scientist
CSIRO's Data61
Australia
Zhenchang.Xing@data61.csiro.au

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Chunyang Chen
Professor
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Xin Peng
Deputy Dean & Professor
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